

Assignment - 2

1. Discuss application firewalls and New Generation Firewall with real world examples.

An application firewall filters traffic for specific apps (after Layer 7), blocking threats like SQL injection or cross-site scripting.

Example: - a bank protects its online portal with a web application firewall (WAF) such as Cloudflare or Akamai.

A Next-Generation Firewall (NGFW) is a security system that not only filters traffic based on IP and ports but also inspects application data, detects threats, and controls user activity in real time.

Example: - A company using an NGFW from Palo Alto Networks can block employees from using risky apps like torrents, detect malware in downloads, and stop hackers trying to access internal systems - all at once.

2. Analyze HTTP / HTTPS security issues like hidden manipulation and HSTS

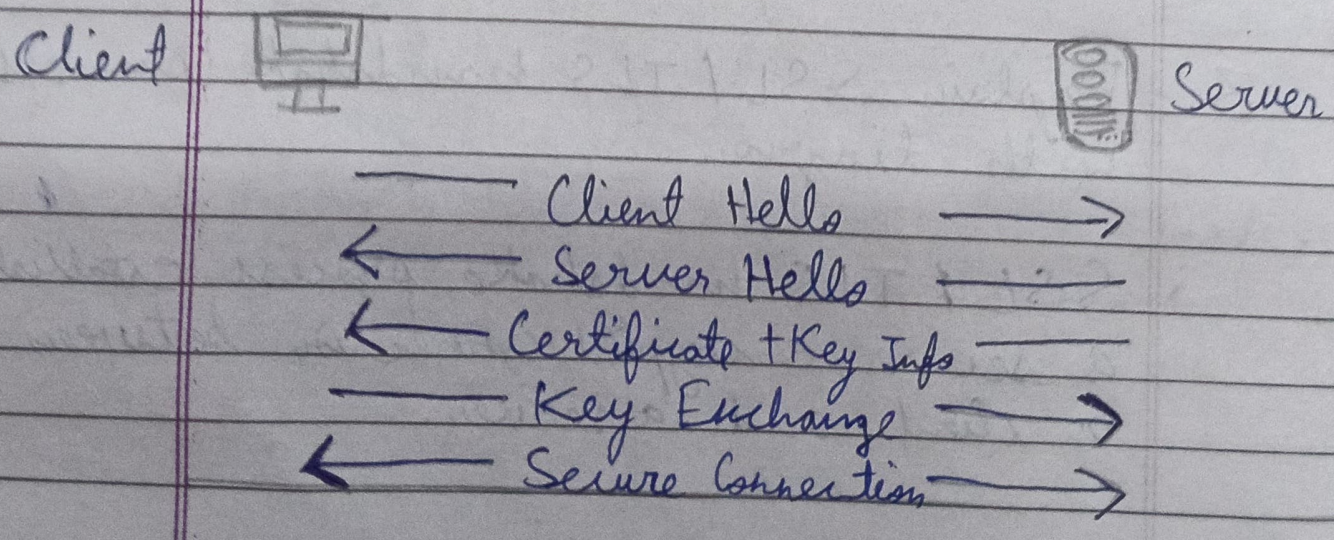
HTTP / HTTPS security issues :-

- Hidden manipulation : In HTTP, data is unencrypted, so attackers can intercept and modify it (e.g., fake login pages on public Wi-Fi).
- HTTPS issues : Safer, but can still be attacked via fake certificates or downgrade attacks (SSL stripping).
- HSTS : A security feature that forces browsers to use HTTPS only. Sites like Google use it to prevent such attacks.

3. Explain SSL / TLS handshake process with diagram.

SSL / TLS handshake process establishes a secure, encrypted connection between a client and a server.

1. Client Hello → Browser sends supported TLS versions, cipher suites and a random number.
2. Server Hello → Server selects settings, sends its digital certificate and its own random number.
3. Certificate verification → Client verifies the server's certificate (trust check).
4. Key Exchange → Client creates ~~as~~ a session key, encrypts it with the server's public key, and sends it.
5. Secure Connection → Both sides use the session key to encrypt communication.



Project

Section -1

Firewall and network architecture case study

- A company suffered data breach despite having a firewall, analyze how packet filtering and stateful firewalls might have fail. Suggest improvements using NGFW and application firewalls.

The breach likely happened because packet filtering firewalls only check IP, port, and protocol, so malicious traffic over allowed ports (like HTTPs) passed through. Stateful firewalls track sessions but still trust established connections, allowing malware or attacks inside normal traffic.

Improvement:

Use a NGFW like Palo Alto Networks to inspect traffic deeply and detect threats, and a Web Application Firewall like Cloudflare to block

application - level attacks (e.g., SQL injection)

Section - 2

Protocol analysis case study

- Analyze an attack involving HTTP hidden manipulation and lack of HSTS. Suggest preventive measures.

The attack used HTTP and lack of HSTS to perform MITM and hidden manipulation, letting attackers intercept and modify data (e.g., fake login pages). Without HSTS, users could be downgraded from HTTPS to HTTP.

Prevention :-

- Enable HSTS (like Google)
- Enforce HTTPS for all traffic
- Use valid TLS certificates
- Deploy WAF (e.g., Cloudflare)